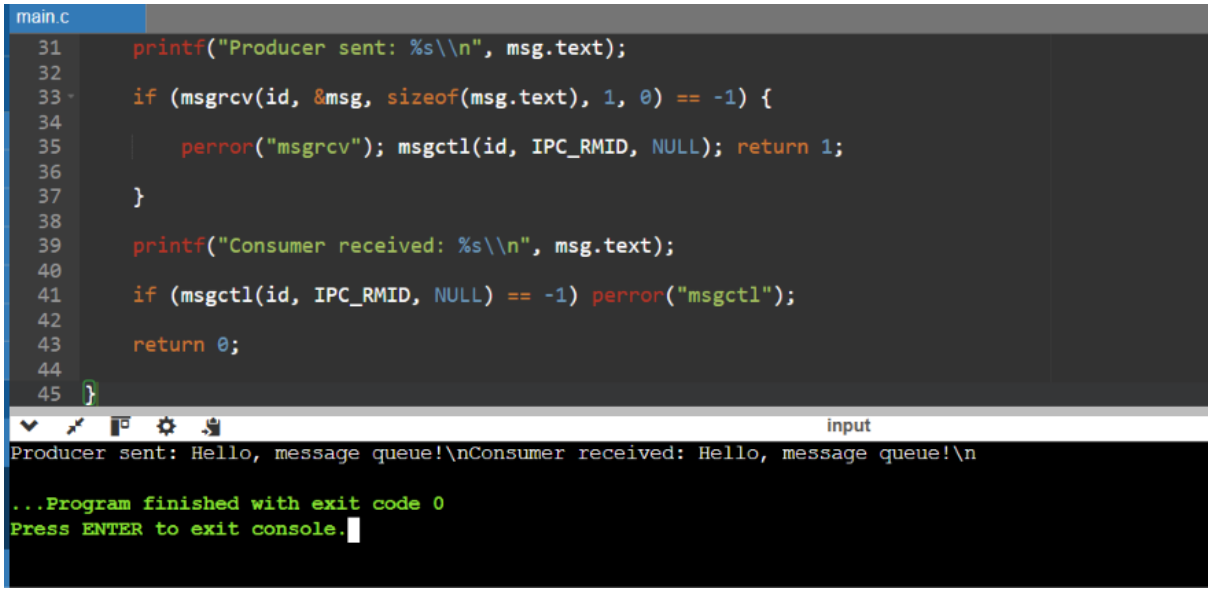
Experiment 10

IPC Using Message Queue

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <sys/ipc.h>
#include <sys/msg.h>
struct message { long type; char text[100]; };
int main(void) {
int id = msgget(IPC_PRIVATE, IPC_CREAT | 0600);
struct message msg;
if (id == -1) { perror("msgget"); return 1; }
msg.type = 1;
strcpy(msg.text, "Hello, message queue!");
if (msgsnd(id, &msg, sizeof(msg.text), 0) == -1) {
perror("msgsnd"); msgctl(id, IPC_RMID, NULL); return 1;
}
printf("Producer sent: %s\\n", msg.text);
if (msgrcv(id, &msg, sizeof(msg.text), 1, 0) == -1) {
perror("msgrcv"); msgctl(id, IPC_RMID, NULL); return 1;
}
printf("Consumer received: %s\\n", msg.text);
if (msgctl(id, IPC_RMID, NULL) == -1) perror("msgctl");
return 0;
}
```

Output:


main.c
printf("Producer sent: %s\\n", msg.text);
if (msgrcv(id, &msg, sizeof(msg.text), 1, 0) == -1) {
perror("msgrcv"); msgctl(id, IPC_RMID, NULL); return 1;
}
printf("Consumer received: %s\\n", msg.text);
if (msgctl(id, IPC_RMID, NULL) == -1) perror("msgctl");
return 0;
}
input
Producer sent: Hello, message queue!\nConsumer received: Hello, message queue!\n
...Program finished with exit code 0
Press ENTER to exit console.